Does paying more help with earning more? Exploring net price and graduates' earnings as US Bachelor's granting colleges.

ECON 3916 Midterm Project · Lily Wang

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Question	Among US colleges that mainly award bachelor's degrees, how much of the variation in median earnings of federal-aid recipients 10 years after entry in the 2009-2011 cohort is associated with net price after accounting for selectivity and share of undergrads receiving Pell grant?
Hypothesis	I expect selectivity, measured by lower admissions rates, to increase the median earnings of federal-aid recipients 10 years post-entry, since higher price might mean more resources.
Headline finding	A school with $1,000 higher net price (in 2022 dollars) has about 0.5% higher median earnings (95% CI [0.3%, 0.7%]). The association grows to 1.1% without controls, but it is a conditional association, not the pure effect of price change.
Data	One row is one college institution from the College Scorecard from the U.S. Department of Education. Earnings: 'Most-Recent-Cohorts-Institution-modified.csv' downloaded October 26th but modified November 11th. Historical data downloaded November 11th and modified the same date – 'MERGED2010_11_PP.csv'. Glossary and cohort information can be found in 'CollegeScorecardDataDictionary.xlsx'. CPI data from FRED used for price conversions is in this file: 'CPIAUCNS.csv'.
Method	OLS of log median earnings on net price ($1,000s, 2022 dollars), admission rate, Pell share and sector, with robust standard errors (HC3).

1. The Question and the Data

Whether a more expensive college pays off is a question families ask when choosing a school and legislators ask when setting tuition and funding. It matters to students and families, to state boards, and to federal regulators who judge schools on outcomes. My question is whether, among US colleges that mainly awarded bachelor's degrees in 2009-10, a higher net price goes with higher median earnings of federal-aid recipients 10 years after entry, holding sector, admission rate and Pell share fixed. The outcome is MD_EARN_WNE_P10, in logs. The main explanatory variable is net price. I originally planned to include instructional spending as

well, but dropped it because tuition revenue funds instruction, so spending plausibly sits between price and earnings and would absorb part of what price does.

The data are a product of measurement decisions. The Scorecard is assembled by the US Department of Education from federal tax and student-aid records and from the colleges' own annual reports. Earnings cover only students who received federal aid and were working and not enrolled, so graduates in further study and non-workers are absent. The figure is a median per institution, pooled over two entry cohorts (academic years 2009-10 and 2010-11) and measured in 2020 and 2021, in 2022 dollars. Net price is the average price after grants for aid recipients, so it describes the same group as the earnings. We used the Pell share and the sector from the 2010-11 file, where they describe AY2009-10 as well as the admission rate from the same file, where it describes Fall 2010. Using the current file would have instead measured price twelve years after the cohort entered which would cause a timing discrepancy. I then converted net price to 2022 dollars with the CPI-U (factor ~1.35) so it is in the same dollars as earnings. The file label is not the year the data describes, which I confirmed in the Scorecard's cohort map 'CollegeScorecardDataDictionary.xlsx'.

Getting from the two files to a usable table took several steps. Earnings load as text because suppressed values are coded as "PS" for privacy suppressed; I counted the 623 of them, then converted them to missing because we have no information on them. Net price sits in two columns, one filled for public schools and one for private, so I combine them. I joined the two files - the current one and the 2010-11 one - on UNITID; of 6,278 schools in the current file, 4,765 (75.9%) appear in the 2010-11 file. I filtered to schools whose predominant degree in 2010 was a bachelor's because that is our target population of interest. Finally I converted dollars with the CPI to 2022 dollars, since the prices in the older file were all nominal prices during that time period.

Several kinds of missing data affect the answer we are looking for. The current file has 6,278 rows and 3,379 columns; the 2010 file 7,470 rows. Among the 1,727 schools that were bachelor's-focused in 2010 and are in the current file, earnings are missing for 53 (3.1%), net price for 111 (6.4%), Pell share for 14 (0.8%), admission rate for 258 (14.9%) and SAT for 457 (26.5%). Earnings are missing partly through privacy suppression and partly because nothing was reported (in the full current file, 623 "PS" values and 362 blanks). Suppression is triggered by small cohorts, not by the earnings value, so I would label this MAR or MNAR given school size and sector. Admission rate is missing mostly where a school has open admission and no rate exists; that is structural and tied to selectivity itself, so effectively MNAR and the model leaves out those schools. SAT is missing for 26.5%, which is tied to test policy and to scores themselves as well as whether schools are open admission or not, which is MNAR. Net price is given originally as either NPT4_PUB for public schools or NPT$_PRIV for private schools, so one

of them will be empty for any school because it will either be public or private. The remaining blanks are likely MAR, either because the school did not report or because the school had no aid recipients.

Table 1. From 68 listed films to the sample

	Films
	6278
Step	
No match in 2010_11 file on UNITID	-1513
In both files but not bachelor's focused in 2010	-3038
No earnings	-53
No net price	-89
No admission rate	-163
Schools in the sample	1422

Table 2. Summary statistics, Bachelor's-focused schools in 2010 file

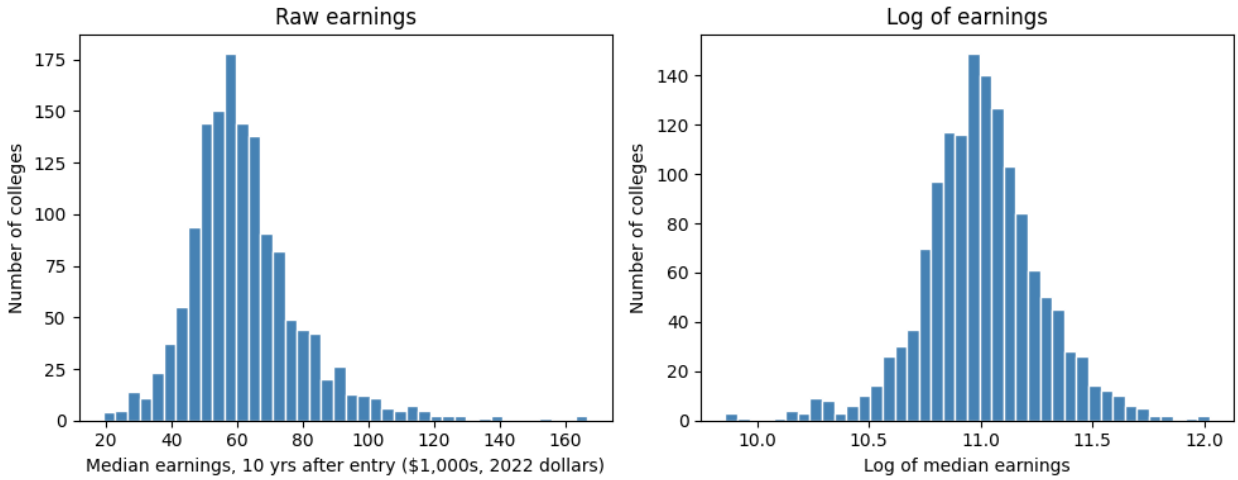
Variable (units)	Mean	Median	SD	Min	Max	N
Median earnings, 10 yrs after entry ($ 2022 dollars)	62191.9	59628.0	17101.2	19092.0	166642.0	57
Net price, AY2009-10 ($ per year)	17267.3	16658.5	7084.4	-20688.0	44950.0	57
Pell share of undergraduates, AY2009-10 (%)	34.1	31.0	16.6	0.0	100.0	57
Admission rate, Fall 2010 (%)	64.8	66.9	19.2	7.2	100.0	57
Degree-seeking undergraduates, Fall 2010 (students)	5246.1	2419.0	6855.7	36.0	56977.0	57
Average SAT score, Fall 2010 (points)	1067.9	1049.5	128.0	715.0	1528.0	53

2. Descriptive Evidence

Earnings are right-skewed The mean is $62,192 and the median $59,628, and the skewness is 1.23, driven by a tail of high-earning schools. The five highest are the California Institute of Technology ($166.6k), Harvey Mudd ($164.2k), MIT ($152.6k), Babson ($139.0k) and Olin ($138.8k), all engineering or business schools that admit between 10% and 40% of applicants.

After taking logs the skewness is −0.14, so I use log earnings as the outcome. A coefficient on net price then reads as an approximate percent difference.

Figure 1



The mean of earnings is skewed right with a mean of $62,192 and a median of $59,628, skewness 1.23.

Figure 2 plots log earnings against net price, split by sector. Schools with a higher net price tend to have higher earnings (r = 0.40), though it is not a strong relationship. The relationship can be seen between sectors as well (public r = 0.46, private nonprofit r = 0.50); it is weaker for the few for-profits (r = 0.26). Public schools cluster at lower net prices and private nonprofits spread across higher ones.

Figure 2

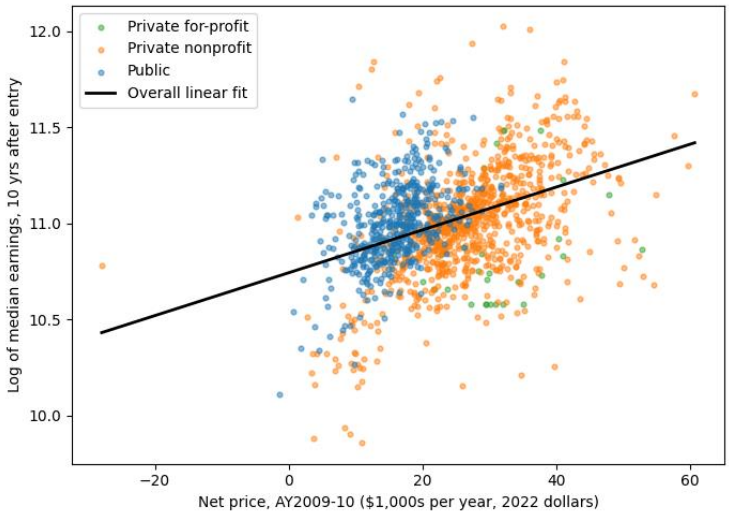
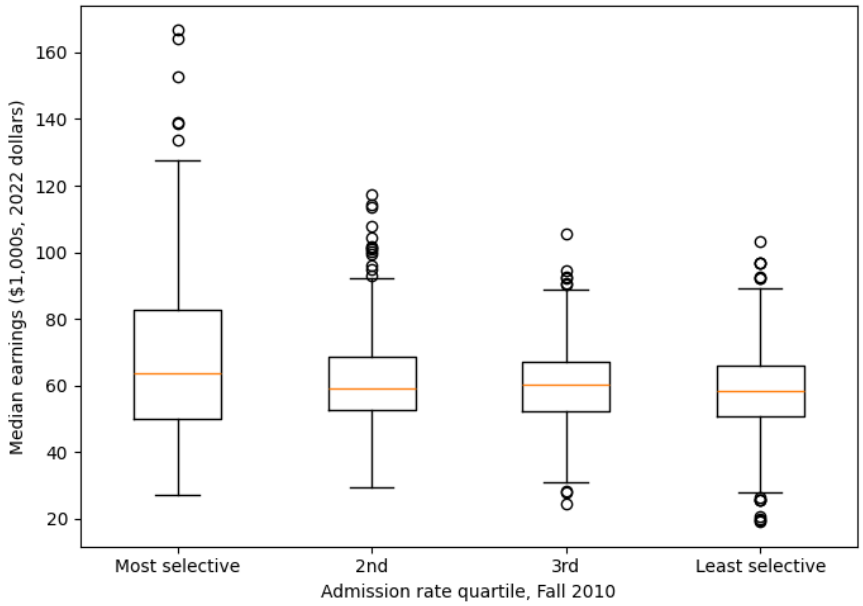


Figure 3 shows why selectivity is a control. Median earnings are $63.8k in the most selective quartile of schools (admission rate under 53%) and $58.3k in the least selective, with $59.2k and $60.3k in between. The most selective quartile has far more spread, with a middle half running from about $50k to above $80k, against roughly $15k for the others. Median net price barely moves across quartiles ($22.4k, $22.7k, $23.6k, $20.8k), and admission rate is almost uncorrelated with net price (−0.06) but negatively correlated with log earnings (−0.24). So selectivity relates to earnings but not to price.

Figure 3



3. The Model

I fit the model by OLS, with heteroskedasticity-robust (HC3) standard errors:

$$\log(\text{Earnings}_i) = \beta_0 + \beta_1 \text{NetPrice}_i + \beta_2 \text{AdmRate}_i + \beta_3 \text{PellShare}_i + \beta_4 \text{Nonprofit}_i + \beta_5 \text{ForProfit}_i + u_i$$

Table A1 in the appendix has the results; column 2 is the main model and column 3 re-estimates it with robust standard errors. Without controls (column 1), a $1,000 higher net price goes with 0.0111 higher log earnings, about 1.1%. Holding sector, selectivity and Pell share fixed (column 2), the coefficient is 0.0050 (classical SE 0.0007; robust SE about 0.0011). A school with $1,000 higher net price, in 2022 dollars, has about 0.50% higher median earnings, roughly

$297 at the median of $59,628. The 95% confidence interval is [0.3%, 0.7%] with robust standard errors (0.35% to 0.64% with classical ones). That is clearly above zero so the direction is not in doubt. The coefficient shrinks by more than half when controls enter, so most of the raw association reflects other differences between schools.

The second coefficient I interpret is Pell share: each 10 percentage points higher Pell share goes with 9.4% lower earnings, holding the other variables fixed ($p < 0.001$). Two other estimates complete the picture. A school that admits 10 percentage points more applicants earns about 2.9% less (coefficient −0.0029 per percentage point). Private nonprofits earn about 6.3% less than public schools (−0.0649, SE 0.0130), while for-profits are 4.5% lower, an estimate too imprecise to distinguish from zero (−0.0461, SE 0.0368 classical, 0.0544 robust). Adding the controls raises R^2 from 0.159 to 0.548 (N = 1,422).

- Sector blocks a confounding path. Public schools charge less and have different earnings, so leaving it out would mix a sector difference into the price coefficient. It would be a bad control only if net price could change a school's sector, which it cannot. The for-profit estimate could be imprecise because there are few of them.

- Pell share blocks a confounding path, since student income affects both aid, and so net price, and later earnings. It would be a bad control if a high price discourages low-income students from enrolling, because Pell share would then be partly caused by price and controlling for it would remove part of the effect I want. Net price and earnings also both describe aid recipients, so there is some overlap by construction.

- Admission rate mostly adds precision. Figure 3 shows it barely moves with net price, so it blocks little confounding, but it explains earnings variation unrelated to price. It would be a bad control if price changes who applies, which would put it on the path from price to earnings. Including it also drops open-admission schools, which have no rate.

- Instructional spending is left out on purpose. Tuition revenue funds instruction, so spending plausibly sits between price and earnings, and controlling for it would absorb part of what price does.

4. Diagnostics

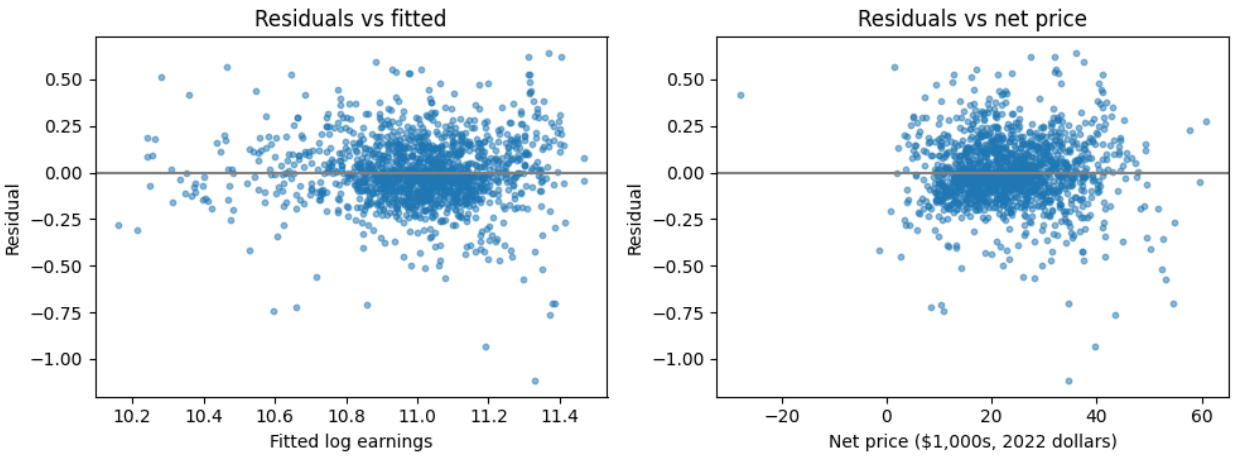
The residual plots (Figure 4) are centered on zero, with no curve against the fitted values or against net price, so a straight line is a reasonable shape. A handful of large negative residuals, between about −0.7 and −1.1, sit at higher fitted values and higher net prices. The Breusch-

Pagan test rejects constant variance ($p < 0.001$). The test is sensitive with 1,422 schools, but the effect on the results is not small: the standard error on net price rises from 0.0007 with classical errors to about 0.0011 with HC3, around 60% larger. The other changes are smaller: Pell share 0.0003 to 0.0004, private nonprofit 0.0130 to 0.0147, and for-profit 0.0368 to 0.0544. I therefore report robust standard errors, and the coefficient on net price stays clearly positive under both.

Collinearity is not a concern. The variance inflation factors are 1.02 for admission rate, 1.23 for for-profit, 1.46 for Pell share, 1.72 for private nonprofit and 2.12 for net price. Net price is the highest because public schools charge less, but all are far below 5.

For influence, the three largest Cook's distances belong to Chamberlain University's Florida (0.069), Illinois (0.059) and Ohio (0.057) campuses, all far below the usual concern levels of 0.5 or 1. Dropping the three leaves the net price coefficient unchanged at 0.005. I keep them because they are real schools and not outliers. They do point to a data issue: campuses of one institution share a single earnings figure so institutions that have multiple campuses and multiple rows are not independent observations. HC3 corrects unequal variance but not this dependence, so the standard errors may be understated.

Figure 4



5. Threats and Conclusion

For the coefficient on net price to be causal, schools with different net prices would have to be alike in everything else that affects graduates' earnings, once sector, admission rate and Pell share are fixed. They are not. The biggest gap is what schools teach: the five highest earners are engineering and business schools, and a school that sends most students into engineering or nursing can charge more and produce higher earners. Nothing in this model sees program mix.

Region is a second omitted factor, since schools in high-wage metros tend to have both higher prices and higher earnings. The variables are also loose stand-ins. Net price describes AY2009-10 for aid recipients, while earnings come from two entry cohorts (2009-10 and 2010-11) measured in 2020-21; a school's price a decade ago only approximates what that cohort paid, and I converted dollars with the CPI. Earnings cover only federal-aid recipients who were working and not enrolled, so graduates who went on to further study are absent. Schools that closed or left the Scorecard before the current file are also missing, so the estimate describes survivors. Finally, several campuses share one earnings figure, which may leave standard errors understated. So the estimate is a conditional association among surviving bachelor's-focused schools, not the effect of raising or cutting a school's price.

Among 1,422 schools, a $1,000 higher net price goes with about 0.50% higher median earnings (95% CI [0.3%, 0.7%]), after accounting for sector, selectivity and Pell share, down from about 1.1% without controls. The estimate is statistically precise, but a reader should trust the direction more than the size, because omitted program mix and regional wages could easily shift it, and I would not read it as a causal return to paying more. With another month I would add each school's mix of majors from the Scorecard's field-of-study file, add a regional wage measure, rerun the model with the 2009-10 and 2011-12 files to test the timing choice and with SAT in place of admission rate, and collapse campuses to one row per institution.

Appendix

Regression Table (Figure 1A)

*Note: an asterisk next to a coefficient means p<0.05

	(1) Simple	(2) w/ Controls	(3) Controls, HC3 robust SE
Constant	10.7425*	11.4548*	11.4548*
	(0.0172)	(0.0276)	(0.0367)
Net price ($1,000s, 2022 dollars)	0.0111*	0.0050*	0.0050*
	(0.0007)	(0.0007)	(0.0011)
Admission rate (percentage points)		-0.0029*	-0.0029*

	(1) Simple	(2) w/ Controls	(3) Controls, HC3 robust SE
		(0.0003)	(0.0003)
Pell share (percentage points)		-0.0099*	-0.0099*
		(0.0003)	(0.0004)
Private nonprofit (vs. public)		-0.0649*	-0.0649*
		(0.0130)	(0.0147)
Private for-profit (vs. public)		-0.0461	-0.0461
		(0.0368)	(0.0544)
N	1,422	1,422	1,422
R-squared	0.159	0.548	0.548